

# Privacy in LLMs: An Introduction

Tanmoy Chakraborty

Professor, IIT Delhi

<https://tanmoychak.com/>



Trustworthy Large Language Models

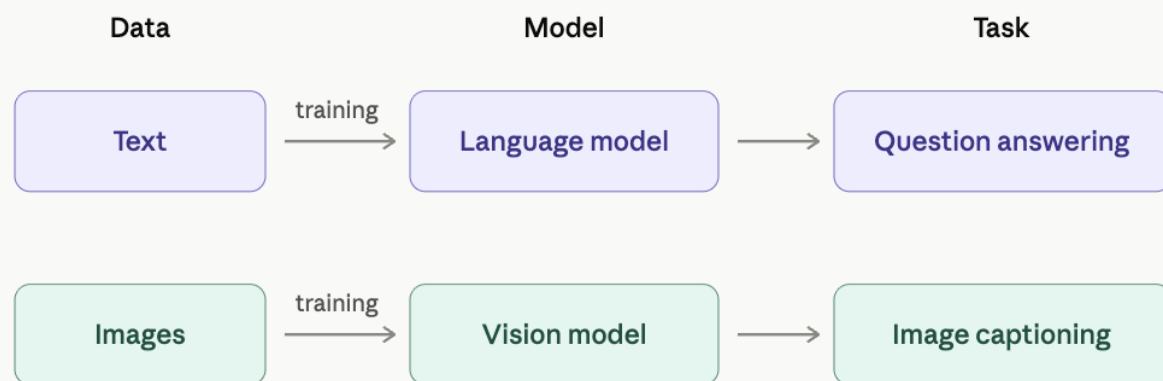


# What is Privacy?

Ability of an individual to seclude themselves or withhold information about themselves, and thereby express themselves selectively.

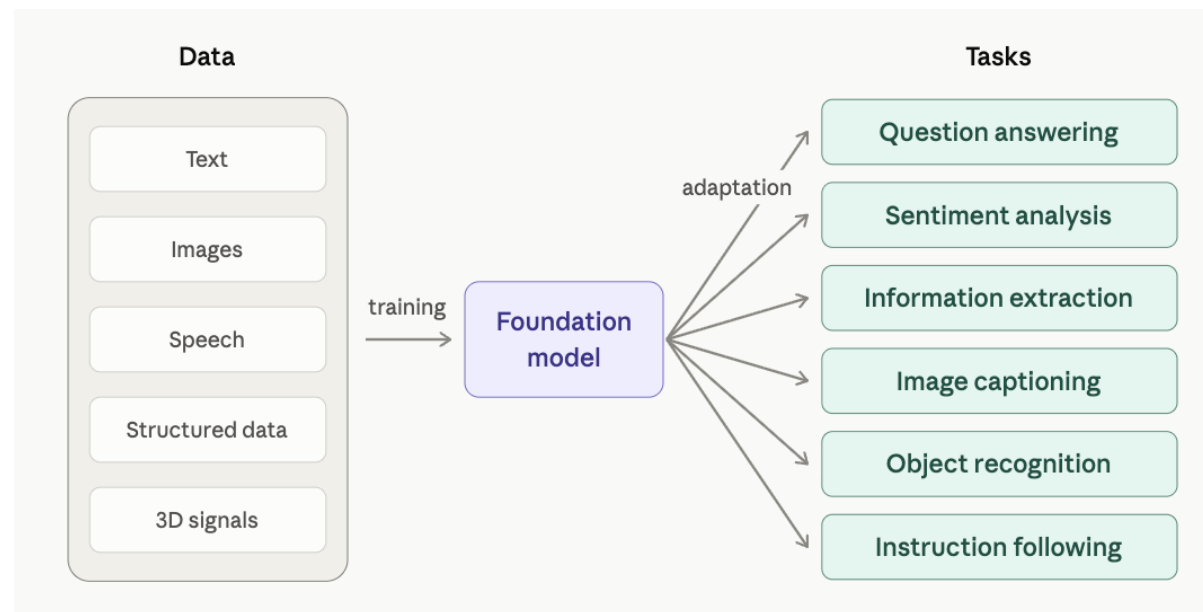
*(“right to be left alone”)*

# Paradigm shift of foundation models



## Before foundation models

Task-specific data and models, trained independently



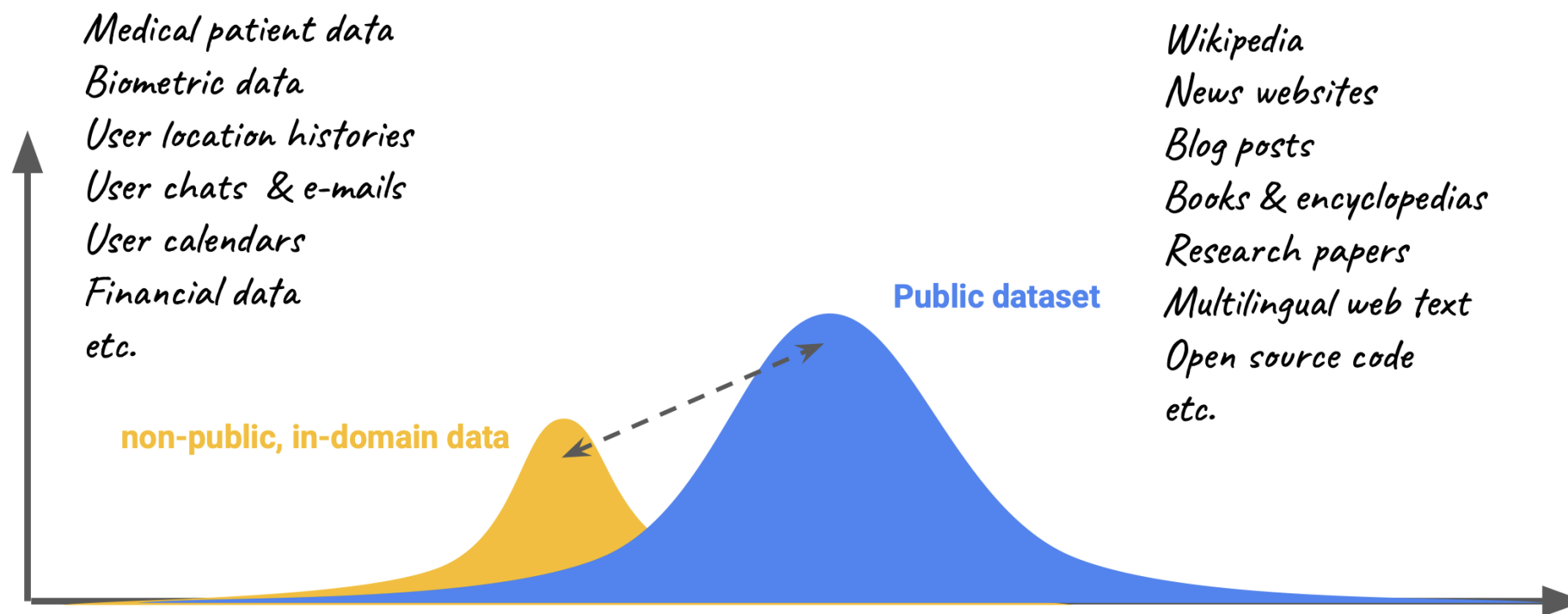
## Post foundation models

Foundation models are trained with data across various modalities, then **adapted to a wide range of downstream tasks**.

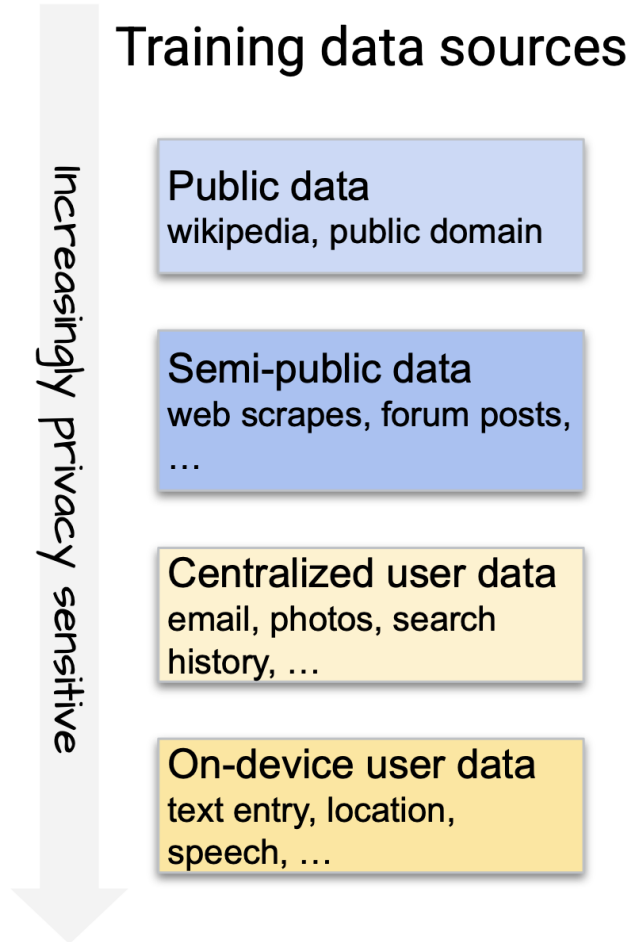
<https://arxiv.org/pdf/2108.07258>



# Foundation models access domain-specific data



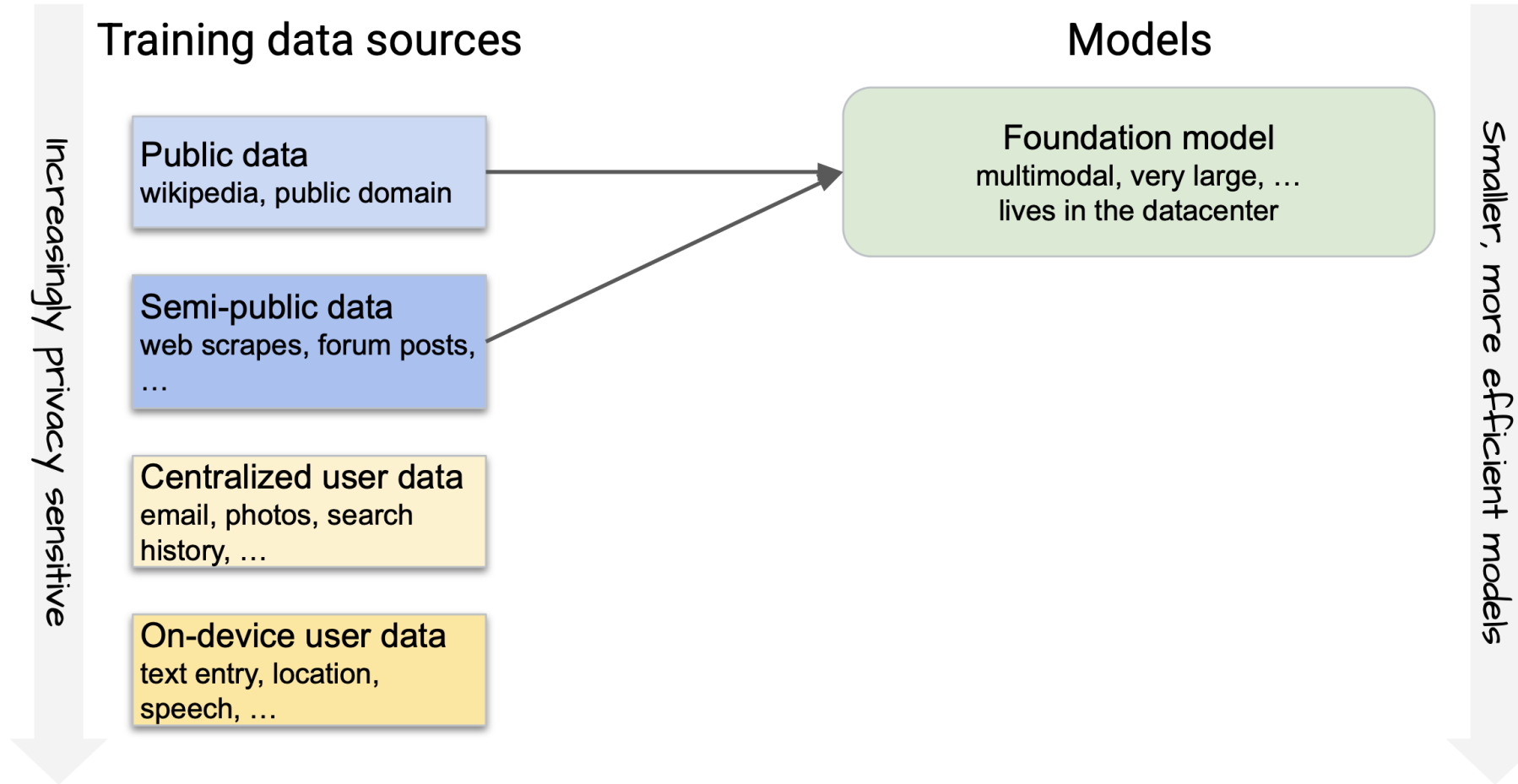
# Incorporating in-domain training data sources



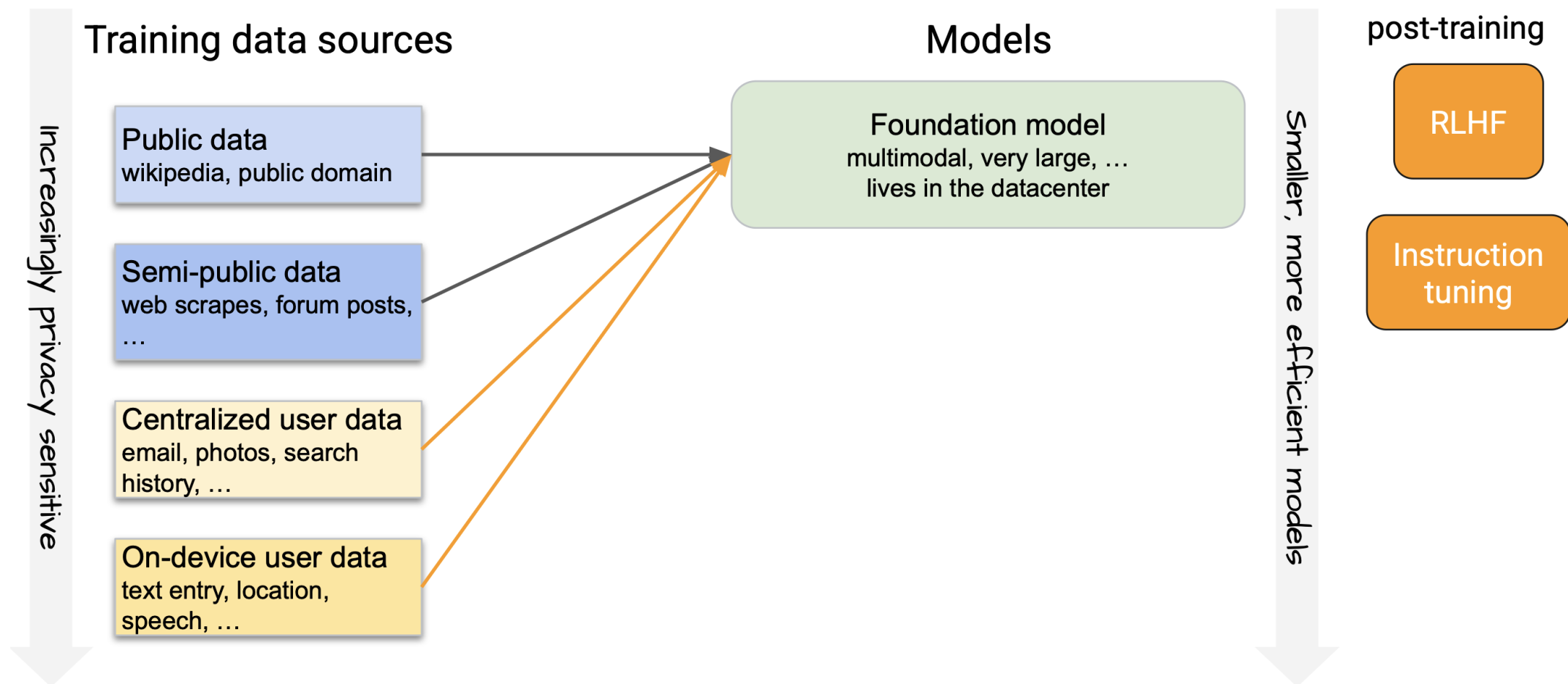
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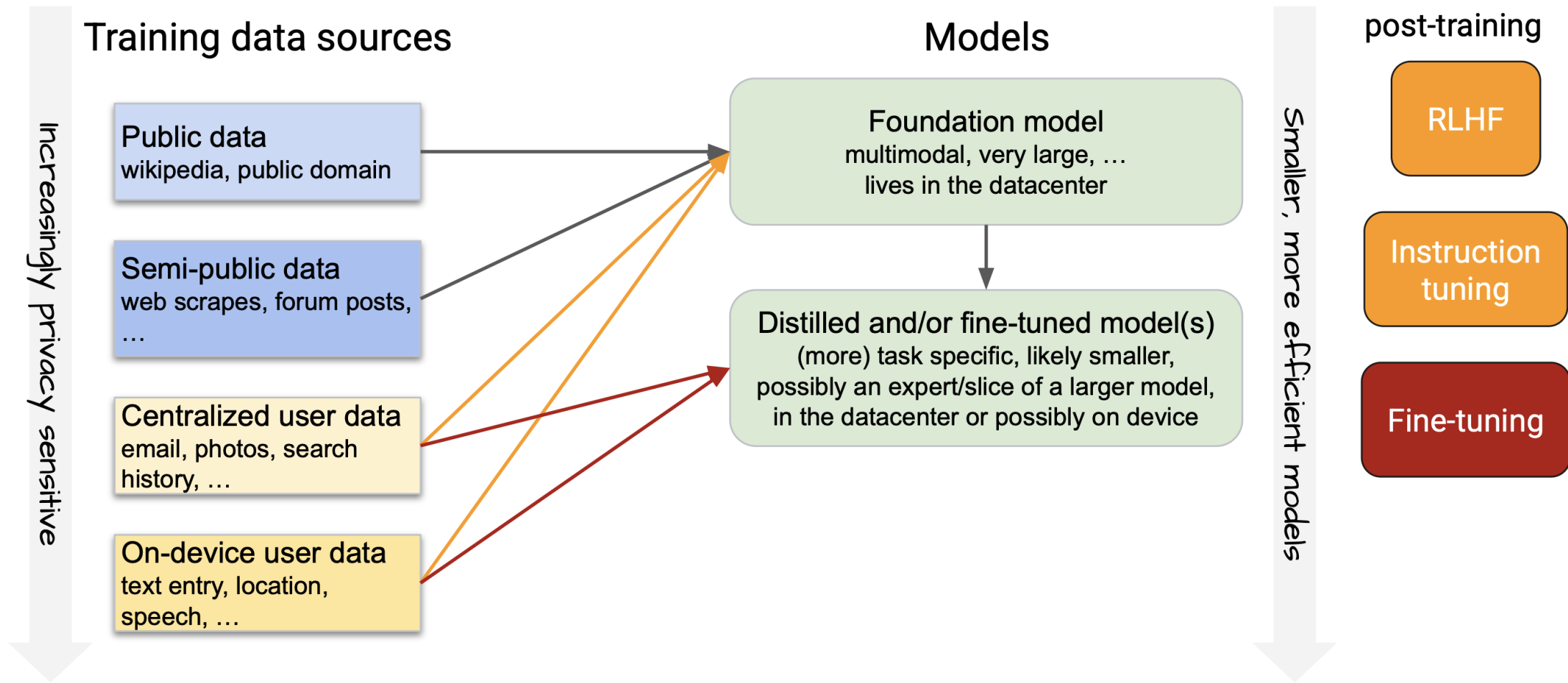
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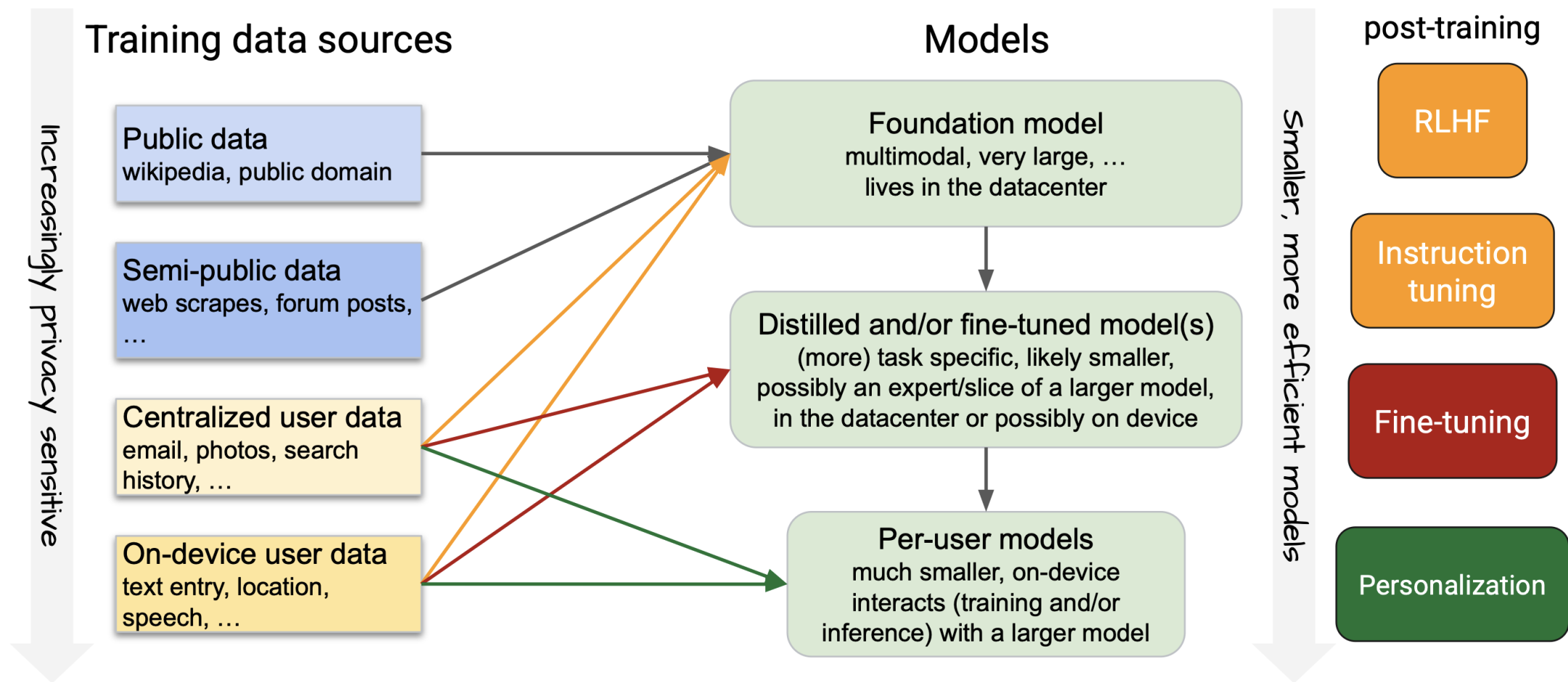


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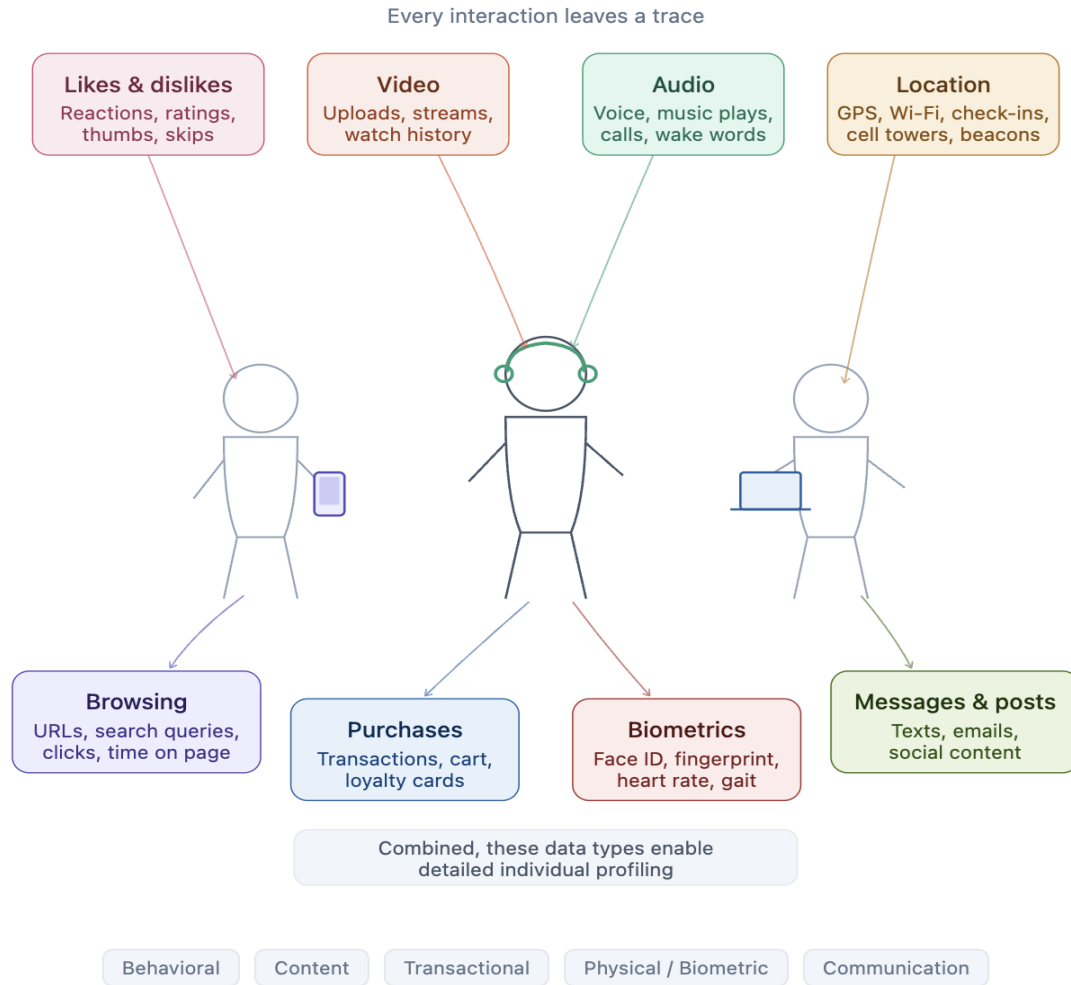
# Incorporating in-domain training data sources



[https://bpb-eu-w2.wpmucdn.com/blogs.bristol.ac.uk/dist/a/1194/files/2025/02/2025-06-16\\_17-Summer-School-Bristol.pdf](https://bpb-eu-w2.wpmucdn.com/blogs.bristol.ac.uk/dist/a/1194/files/2025/02/2025-06-16_17-Summer-School-Bristol.pdf)



# People generate data about themselves



## Different forms of Data

**Behavioural data:** Likes/Dislikes, Purchases

**Identifiable Data:** Name, Address

**Sensitive Data:** Biometric information, medical information.

All of it could be present in an LLM training corpus, tokenized and absorbed into a model's knowledge after training, rather than just being stored in a database.

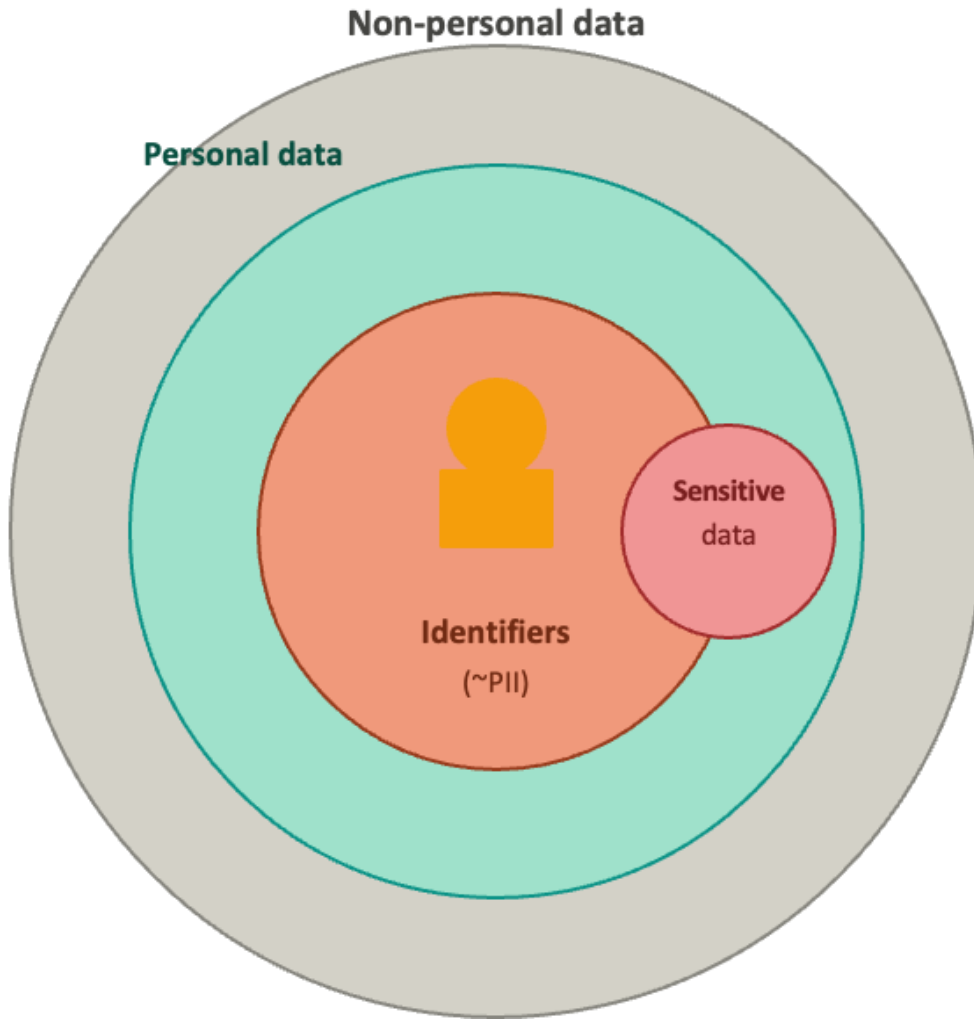


# Same data, two uses

| Data type                           | Beneficial use                                | Potential misuse                             |
|-------------------------------------|-----------------------------------------------|----------------------------------------------|
| <b>Location data</b>                | Navigation, disaster response, urban planning | Tracking journalists, person surveillance    |
| <b>Health &amp; biometric data</b>  | Early disease detection, clinical research    | Insurance discrimination, employer screening |
| <b>Behavioral &amp; social data</b> | Fraud detection, public health modelling      | Voter manipulation, social credit scoring    |



# Data classification



## Non-personal data

Cannot be linked to any individual, even indirectly. Safe to share and publish without restriction -- weather readings, open statistics.

## Personal data

Any information relating to an identified or identifiable individual, directly or indirectly. Cannot be freely shared or published; processing requires a legal basis (e.g., consent, legitimate interest) -- IP address, email etc.

## Identifiers: PII

Attributes that identify a specific individual on their own, either directly or because they are so distinctive they single out exactly one person -- SSN, passport number.

## Sensitive data

A special category requiring heightened protection. Processing requires explicit consent or legal basis -- health, racial origin, religion, sexual orientation, biometrics.



# Privacy regulations on AI and LLMs

| Regulation                                                  | Introduced In                            | Applies To                                                                                                   | Why it matters                                                                                                     |
|-------------------------------------------------------------|------------------------------------------|--------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------|
| General Data Protection Regulation (GDPR)                   | 2018                                     | Processing of personal data of individuals in the EU, regardless of where the company is located             | Establishes rights such as consent, access, deletion, and restrictions on profiling and automated decision-making. |
| Digital Personal Data Protection Act (DPDP)                 | 2023                                     | Processing of digital personal data in India and certain extraterritorial processing involving Indian users  | Emphasizes consent, purpose limitation, data security, and user rights.                                            |
| Health Insurance Portability and Accountability Act (HIPPA) | 1996<br>(Privacy Rules enforced in 2003) | Protected Health Information (PHI) handled by covered entities.                                              | Imposes strict safeguards on healthcare data and limits its disclosure and use.                                    |
| EU AI Act                                                   | 2024                                     | AI systems placed on the EU market, with additional obligations for high-risk and general-purpose AI models. | Introduces transparency, risk management, and governance requirements specifically for AI systems.                 |



# Privacy principles challenged by LLMs

| Privacy Principle                             | Legal Basis                                      | The LLM Challenge                                                                                                                                  |
|-----------------------------------------------|--------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------|
| Consent & Purpose Limitation                  | GDPR Articles 6, 7;<br>DPDP Article Sections 4-6 | <b>Can we legally train on this data and for this particular purpose?</b><br>Large web corpora often lack clear consent and purpose specification. |
| Right to Erasure<br>("Right to be Forgotten") | GDPR Art. 17;<br>DPDP Section 12(3)              | <b>Can we make a model forget one individual after training?</b><br>Machine unlearning remains an unsolved problem at scale.                       |
| De-identification &<br>Re-identification Risk | HIPAA Safe Harbor<br>Section 164.514             | <b>Is redaction enough?</b><br>Generative models may reconstruct or re-identify supposedly anonymized information.                                 |
| Transparency & Data<br>Provenance             | EU AI Act Article 53(1)(d)                       | <b>Can we explain what the model was trained on?</b><br>Internet-scale datasets make training-data documentation difficult.                        |



# Compliance varies across providers

| Category | Keyword          | Requirement (summarized)                                                                                                            |
|----------|------------------|-------------------------------------------------------------------------------------------------------------------------------------|
| Data     | Data sources     | Describe data sources used to train the foundation model.                                                                           |
|          | Data governance  | Use data that is subject to data governance measures (suitability, bias, and appropriate mitigation) to train the foundation model. |
|          | Copyrighted data | Summarize copyrighted data used to train the foundation model.                                                                      |

## Grading Foundation Model Providers' Compliance with the Draft EU AI Act

Source: Stanford Center for Research on Foundation Models (CRFM), Institute for Human-Centered Artificial Intelligence (HAI)



<https://crfm.stanford.edu/2023/06/15/eu-ai-act.html>

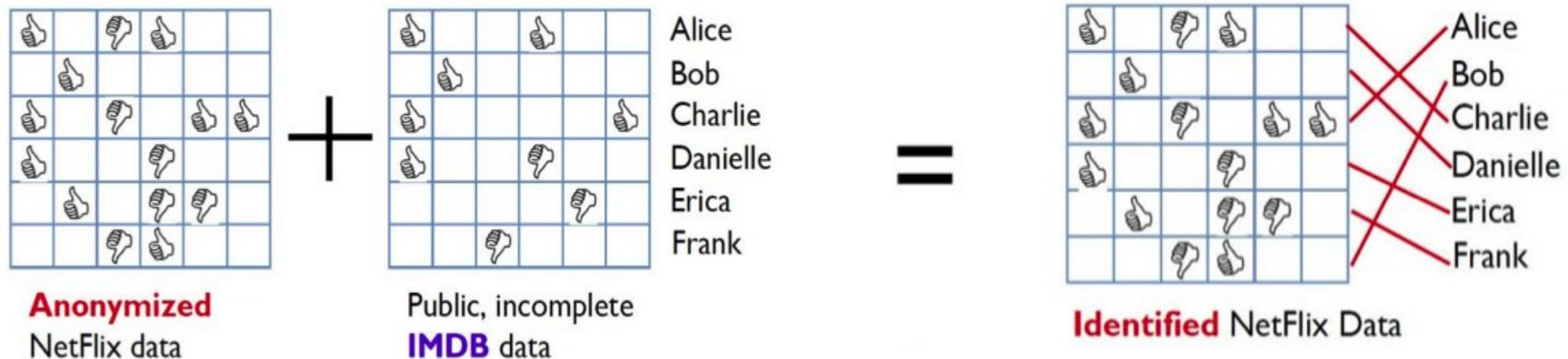




# Real-world privacy failures

Netflix released anonymized data in 2006 for a public contest to find algorithms that could improve their movie recommendation algorithm -- 100,240,507 ratings on a 1-to-5 star scale across 17,770 movie titles

**“Anonymized” data can be re-identified:** Narayanan and Shmatikov. [IEEE S&P '08] demonstrated that anonymized Netflix user data can be linked to publicly available IMDb data, thus undoing the anonymization.



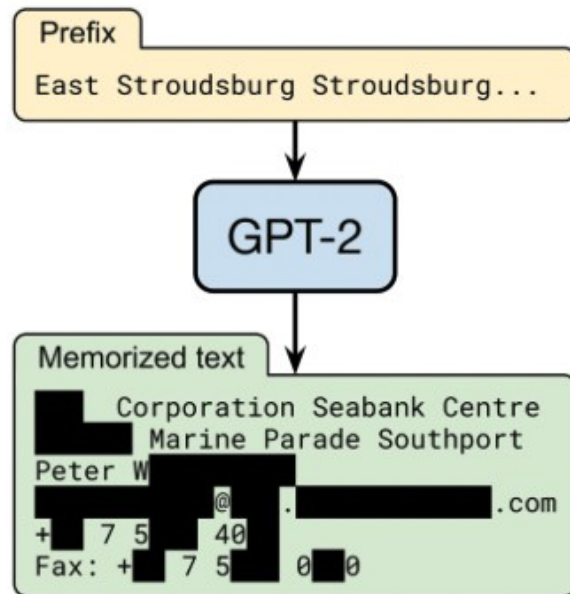
[https://www.cs.cornell.edu/~shmat/shmat\\_oak08netflix.pdf](https://www.cs.cornell.edu/~shmat/shmat_oak08netflix.pdf)





# Real-world privacy failures

- **LLMs memorize their training data:** [Carlini et al. \[USENIX '21\]](#) recovered phone numbers, URLs, and credit-card numbers verbatim from GPT-2 via prompting.



In our [recent paper](#), we evaluate how large language models *memorize* and *regurgitate* such rare snippets of their training data. **We focus on GPT-2 and find that at least 0.1% of its text generations (a very conservative estimate) contain long verbatim strings that are “copy-pasted” from a document in its training set.**

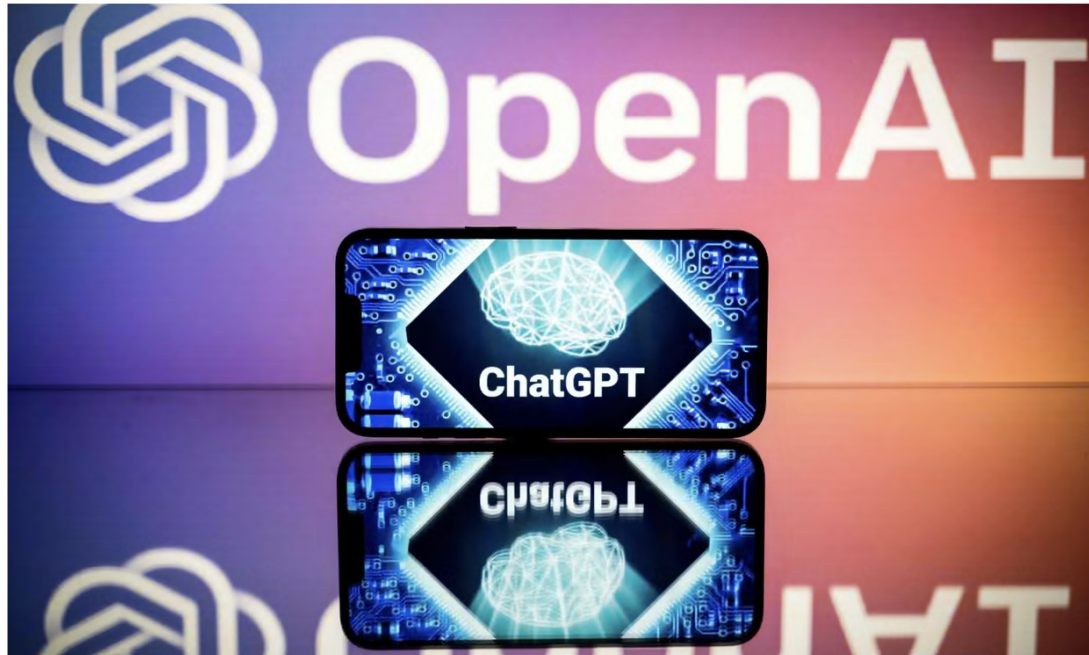
<https://arxiv.org/pdf/2012.07805>



# Consequences of privacy failures

## Italy's privacy watchdog bans ChatGPT over data breach concerns

Measure is in place 'until ChatGPT respects privacy', says Italian Data Protection Authority



*Dan Milmo and agencies*

Sat 1 Apr 2023 08.43 BST

The Italian Data Protection Authority (DPA), the “Garante”, has issued a €15 million fine against OpenAI due to ChatGPT’s alleged **GDPR** compliance failures.

The Garante found that OpenAI violated the following GDPR provisions:

- The GDPR’s principle of “**lawfulness, fairness, and transparency**” at Article 5 (1) (a) of the GDPR
- The **transparency obligations** at Articles 12 and 13
- The **general compliance obligation**, set out at Article 24
- The doctrine of “**data protection by design and by default**” at Article 25
- Certain aspects of the **data breach reporting** obligation at Article 33 (1)
- The requirement to **cooperate with the DPA** at Article 83 (5) (e)

<https://www.theguardian.com/technology/2023/mar/31/italy-privacy-watchdog-bans-chatgpt-over-data-breach-concerns>



# Classical anonymization wasn't built for text

- For decades, privacy meant anonymizing data stored in tables, and three methods were very commonly used.
- These methods remove **identifiers** and suppress/generalize **quasi-identifiers**: fields like age, ZIP-code, gender; characteristics that can identify a person when combined, to protect a sensitive attribute.
  1. **k-anonymity**: blend each record into a group of at least  $k$  indistinguishable persons
  2. **l-diversity**: ensure each group holds  $\geq l$  distinct sensitive values
  3. **t-closeness**: ensure each group's distribution is close to the entire dataset



# K-anonymity

K-anonymity ensures that each individual in a dataset cannot be distinguished from at least  $k-1$  other individuals with respect to the quasi-identifiers in the dataset.

| Original dataset |     |        |           |                 |
|------------------|-----|--------|-----------|-----------------|
| Nr               | Age | Sex    | City      | Disease         |
| 1                | 16  | Male   | Rotterdam | Viral infection |
| 2                | 18  | Male   | Rotterdam | Heart-related   |
| 3                | 19  | Male   | Rotterdam | Cancer          |
| 4                | 22  | Female | Rotterdam | Viral infection |
| 5                | 22  | Male   | Zwolle    | No illness      |
| 6                | 23  | Male   | Zwolle    | Tuberculosis    |
| 7                | 24  | Male   | Zwolle    | Heart-related   |
| 8                | 25  | Female | Utrecht   | Cancer          |
| 9                | 26  | Female | Rotterdam | Heart-related   |
| 10               | 28  | Female | Utrecht   | Tuberculosis    |

<https://utrechtuniversity.github.io/dataprivacyhandbook/k-l-t-anonymity.html>



# K-anonymity

K-anonymity ensures that each individual in a dataset cannot be distinguished from at least  $k-1$  other individuals with respect to the quasi-identifiers in the dataset.

| Original dataset |     |        |           |                 | 2-anonymous dataset |       |        |           |                 |
|------------------|-----|--------|-----------|-----------------|---------------------|-------|--------|-----------|-----------------|
| Nr               | Age | Sex    | City      | Disease         | Nr                  | Age   | Sex    | City      | Disease         |
| 1                | 16  | Male   | Rotterdam | Viral infection | 1                   | =< 20 | Male   | Rotterdam | Viral infection |
| 2                | 18  | Male   | Rotterdam | Heart-related   | 2                   | =< 20 | Male   | Rotterdam | Heart-related   |
| 3                | 19  | Male   | Rotterdam | Cancer          | 3                   | =< 20 | Male   | Rotterdam | Cancer          |
| 4                | 22  | Female | Rotterdam | Viral infection | 4                   | 20-30 | Female | Rotterdam | Viral infection |
| 5                | 22  | Male   | Zwolle    | No illness      | 5                   | 20-30 | Male   | Zwolle    | No illness      |
| 6                | 23  | Male   | Zwolle    | Tuberculosis    | 6                   | 20-30 | Male   | Zwolle    | Tuberculosis    |
| 7                | 24  | Male   | Zwolle    | Heart-related   | 7                   | 20-30 | Male   | Zwolle    | Heart-related   |
| 8                | 25  | Female | Utrecht   | Cancer          | 8                   | 20-30 | Female | Utrecht   | Cancer          |
| 9                | 26  | Female | Rotterdam | Heart-related   | 9                   | 20-30 | Female | Rotterdam | Heart-related   |
| 10               | 28  | Female | Utrecht   | Tuberculosis    | 10                  | 20-30 | Female | Utrecht   | Tuberculosis    |

Colours indicate an 'equivalence class' of quasi-identifiers

<https://utrechtuniversity.github.io/dataprivacyhandbook/k-l-t-anonymity.html>



# L-diversity

L-diversity proposes that there should be at least L different values for the sensitive attribute per combination of quasi-identifiers

## Inference:

- Any female from Rotterdam between 20-30 who participated had a viral infection.
- If you know that your 25-year old female neighbour from Utrecht participated in this study, you learn that she suffers from cancer.

| 2-anonymous dataset |       |        |           |                 |
|---------------------|-------|--------|-----------|-----------------|
| Nr                  | Age   | Sex    | City      | Disease         |
| 1                   | =< 20 | Male   | Rotterdam | Viral infection |
| 2                   | =< 20 | Male   | Rotterdam | Heart-related   |
| 3                   | =< 20 | Male   | Rotterdam | Cancer          |
| 4                   | 20-30 | Female | Rotterdam | Viral infection |
| 5                   | 20-30 | Male   | Zwolle    | No illness      |
| 6                   | 20-30 | Male   | Zwolle    | Tuberculosis    |
| 7                   | 20-30 | Male   | Zwolle    | Heart-related   |
| 8                   | 20-30 | Female | Utrecht   | Cancer          |
| 9                   | 20-30 | Female | Rotterdam | Viral infection |
| 10                  | 20-30 | Female | Utrecht   | Cancer          |

Colours indicate an 'equivalence class' of quasi-identifiers

| 2-anonymous 2-diverse dataset |       |        |           |                 |
|-------------------------------|-------|--------|-----------|-----------------|
| Nr                            | Age   | Sex    | City      | Disease         |
| 1                             | =< 20 | Male   | Rotterdam | Viral infection |
| 2                             | =< 20 | Male   | Rotterdam | Heart-related   |
| 3                             | =< 20 | Male   | Rotterdam | Cancer          |
| 4                             | 20-30 | Female | •         | Viral infection |
| 5                             | 20-30 | Male   | Zwolle    | No illness      |
| 6                             | 20-30 | Male   | Zwolle    | Tuberculosis    |
| 7                             | 20-30 | Male   | Zwolle    | Heart-related   |
| 8                             | 20-30 | Female | •         | Cancer          |
| 9                             | 20-30 | Female | •         | Viral infection |
| 10                            | 20-30 | Female | •         | Cancer          |

Colours indicate an 'equivalence class' of quasi-identifiers and sensitive attributes

if we choose an L of 2, that means that for each combination of Age, Sex and City, there are at least 2 distinct diseases.

<https://utrechtuniversity.github.io/dataprivacyhandbook/k-l-t-anonymity.html>



# T-closeness

A dataset satisfies t-closeness if the **distribution of sensitive attribute values** in every equivalence class differs from the overall dataset distribution by no more than a threshold  $t$ .

2-anonymous 2-diverse dataset

| Nr | Age   | Sex    | City      | Disease         |
|----|-------|--------|-----------|-----------------|
| 1  | =< 20 | Male   | Rotterdam | Viral infection |
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| 9  | 20-30 | Female | •         | Viral infection |
| 10 | 20-30 | Female | •         | Cancer          |

Colours indicate an 'equivalence class' of quasi-identifiers and sensitive attributes

| Disease         | Overall | Green Equivalence Class |
|-----------------|---------|-------------------------|
| Viral Infection | 30%     | 50%                     |
| Cancer          | 20%     | 50%                     |
| Heart-related   | 20%     | 0%                      |
| No illness      | 10%     | 0%                      |
| Tuberculosis    | 10%     | 0%                      |

Earth Mover Distance of distributions:

$$\begin{aligned}
 \text{EMD}(P, Q) &= \frac{1}{2} \sum_i |P_i - Q_i| \\
 &= \frac{1}{2} (|0.5 - 0.3| + |0.5 - 0.3| + |0 - 0.2| + |0 - 0.1| + |0 - 0.1|) \\
 &= \frac{1}{2} (0.2 + 0.2 + 0.2 + 0.1 + 0.1) \\
 &= \frac{0.8}{2} \\
 &= 0.4
 \end{aligned}$$

So, at  
 $t \geq 0.4$ , t-closeness is satisfied  
 $t < 0.4$ , t-closeness is not satisfied





# LLMs break assumptions made by classical privacy

In a table, private data resides in fields that can be pointed to (e.g., a column, a row, a specific cell) so it can be anonymized there.

But with LLMs, the data is spread across multiple new surfaces:

- The **unstructured training text** they are trained on (e.g., web pages, forum posts, books)
- The **weights** that memorize the training data.
- The **prompts** users send at inference time.





# PII Based Text Anonymization

Basic techniques and challenges

# Text anonymization through PII detection

**Personally Identifiable Information (PII)** detection involves detecting specific classes of information like name, location, dates of birth, credit card number, etc. using Named Entity Recognition. Following this PIs might be redacted or replaced with placeholders like [NAME], [LOCATION].

## Original (PII visible)

Patient: John Smith, DOB: 14/03/1985  
Address: 12 Oak Lane, Berlin 10115  
Diagnosis: Type 2 Diabetes  
Attending: Dr. Anna Mueller  
Phone: +49 30 1234567  
Email: j.smith@email.de  
Notes: Stress at Deutsche Bank.

NER  
=> Mask

## After PII Masking

Patient: [PERSON], DOB: [DATE]  
Address: [ADDRESS], [LOCATION] [ZIP]  
Diagnosis: Type 2 Diabetes  
Attending: [PERSON]  
Phone: [PHONE]  
Email: [EMAIL]  
Notes: Stress at [ORG].



# Common PIs

| Entity Type      | Example                                | Tag             |
|------------------|----------------------------------------|-----------------|
| Person Name      | John Smith                             | [PERSON]        |
| Organization     | IIT                                    | [ORG]           |
| Location/Address | Delhi, India                           | [LOC]/[ADDRESS] |
| Date/Time        | 10/05/2026, Sunday                     | [DATE]          |
| Phone/Email      | +919933668800,<br>john.smith@gmail.com | [PHONE]/[EMAIL] |



# Issues with PII based text anonymization

**Problem 1:** Text becomes unnatural and loses utility

"Sarah went to Berlin on March 3rd to meet Tom at Siemens. She said the March 3rd meeting with Tom at Siemens was productive."

Applying PII  
anonymization



"[PERSON] went to [LOCATION] on [DATE] to meet [PERSON] at [ORG]. [PERSON] said the [DATE] meeting with [PERSON] at [ORG] was productive."

The text is now unreadable, has lost context, performance on downstream NLP tasks like summarization, translation and sentiment analysis will degrade on the anonymized text.



# Issues with PII based text anonymization

## Problem 2: Residual Profiling Risk

Even after masking all named entities, text retains:

- Writing style and vocabulary
- Domain specific jargon
- Implicit locations
- Sentiment and opinion patterns

"[PERSON] caught the **local** during rush hour, grabbed a **vada pav** from the corner stall, and got stuck when the monsoon **flooded the tracks** again. Later, [PERSON] graded **semester papers** before the **submission deadline**."

- Local / vada pav / flooded tracks -> **Mumbai**
- grading papers / deadline -> **university academic**

**Every named entity is masked, yet location and occupation still leaked**



# Text Sanitization

Moving beyond PII detection

# Redaction vs Anonymization vs Sanitization

- **Redaction:** Removes the sensitive spans entirely, leaving blanks.

“John lives in Delhi” -> “[REDACTED] lives in [REDACTED]”

- **PII Based Anonymization:** Replace each identifying identity with a typed placeholder

“John lives in Delhi” -> “[PERSON] lives in [LOC]”

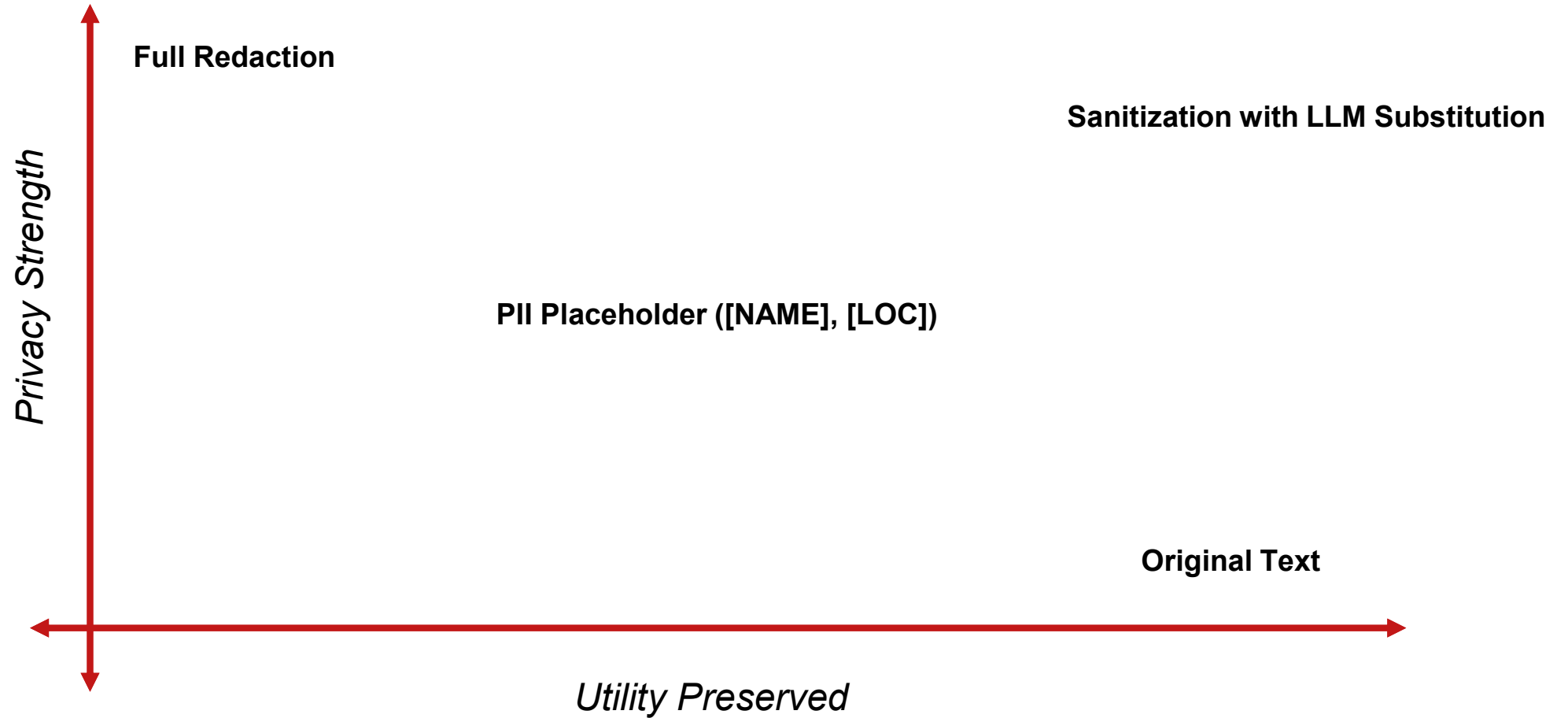
- **Sanitization:** Transforms sensitive content while keeping the text useful for downstream tasks.

“John lives in Delhi” -> “Michael lives in Mumbai”

Sanitization is the umbrella goal: private and usable text. A sanitizer that destroys utility is just an expensive redactor.



# Privacy-Utility Trade-off in Text





# Text sanitization with LLMs: ZSTS (Zero-shot text sanitization)

- **Idea:** a pre-trained LLM already knows which words are unusual in a particular context.
- **Unusual  $\Rightarrow$  Informative  $\Rightarrow$  Sensitive**
- [Albanese et al. 2023] suggested using masked language modelling (MLM) to estimate the probability of each word in a document.
- Words that have probability below a certain privacy threshold  $p$  should be filtered out

<https://arxiv.org/pdf/2311.10785>



# ZSTS: Method

- 1) **Mask:** each word and use BERT to estimate their probabilities, given the context.
- 2) **Redact:** words whose probability falls below a threshold  $p$
- 3) **Generate:**
  - Numeric tokens are replaced with randomly generated number of same length.
  - For a non-numeric token, select BERT's top- $n$  tokens based on probability, retain the top- $k$  candidates within embedding distance  $s$ , and randomly select one for replacement; if fewer than  $k$  candidates satisfy the distance criterion, remove the term entirely.

**Zero-shot method:** no finetuning, no entity lists, no domain-specific training data. Works on medical, legal, dialogue, or social text out of the box.  
*Configurable threshold  $p$  directly controls the privacy-utility tradeoff.*



# ZSTS: Example

**Input sentence:** “My name is John Smith”

**BERT-estimated word probabilities (in context):**

| Word  | P(word)      |
|-------|--------------|
| My    | 0.03         |
| name  | 0.07         |
| is    | 0.06         |
| John  | <b>0.004</b> |
| Smith | <b>0.001</b> |

**$p = 0.01$**

**Step 1) Redact:** “John” and “Smith” fall below the threshold  $p$  and are flagged.

**Step 2) Substitute:** BERT’s top-ranked candidates by embedding similarity:

- John -> David
- Smith -> Williams

**Output sentence:** “My name is David Williams”

*Grammatical flow, meaning and downstream utility are preserved.*



# ZSTS: Results

## Baselines:

- **Microsoft Presidio:** Open-source PII detection and redaction framework combining rule-based patterns and NER models for sensitive information identification.
- **Google Data Loss Prevention:** Production-grade de-identification service that detects PII using pattern matching, ML, checksums, and contextual analysis.

**Dataset:** Action-Based Conversation Dataset, over 10k human-to-human dialogues

**PIIs considered:** *customer name, username, email, phone number, account id, order id, street address, and zip code.*



# ZSTS - Results

| Model                 | Recall | Precision | F1   |
|-----------------------|--------|-----------|------|
| Microsoft Presidio    | 0.56   | 0.30      | 0.39 |
| Google DLP            | 0.65   | 0.66      | 0.66 |
| ZSTS ( $p=1E^{-25}$ ) | 0.06   | 0.81      | 0.10 |
| ZSTS ( $p=1E^{-15}$ ) | 0.26   | 0.80      | 0.40 |
| ZSTS ( $p=1E^{-5}$ )  | 0.54   | 0.29      | 0.37 |
| ZSTS ( $p=1E^{-4}$ )  | 0.68   | 0.21      | 0.32 |
| ZSTS ( $p=1E^{-3}$ )  | 0.84   | 0.15      | 0.26 |
| ZSTS ( $p=1E^{-2}$ )  | 0.94   | 0.10      | 0.19 |
| ZSTS ( $p=1E^{-1}$ )  | 0.98   | 0.06      | 0.12 |

Recall – proportion of unsafe terms properly redacted

Precision – proportion of unsafe terms redacted over all redacted terms

$p$  controls the trade-off between Recall and Precision



# ZSTS: Performance on downstream tasks

| Method                        | Privacy<br>[Recall] | Sentiment<br>[Accuracy] | Topic distance<br>[mean $\pm$ std] | Q&A<br>[Accuracy] | Q&A true<br>[Accuracy] |
|-------------------------------|---------------------|-------------------------|------------------------------------|-------------------|------------------------|
| Presidio                      | 0.56                | 0.999                   | 0.001 $\pm$ 0.001                  | 0.61              | 0.52                   |
| DLP                           | 0.65                | 1                       | 0.0003 $\pm$ 0.0006                | 0.52              | 0.36                   |
| ZSTS Redaction                | 0.98                | 0.997                   | 0.022 $\pm$ 0.021                  | 0.57              | 0.26                   |
| ZSTS Redaction & Substitution | 0.98                | 0.993                   | 0.023 $\pm$ 0.021                  | 0.82              | 0.75                   |

Downstream task utility is maintained along with privacy



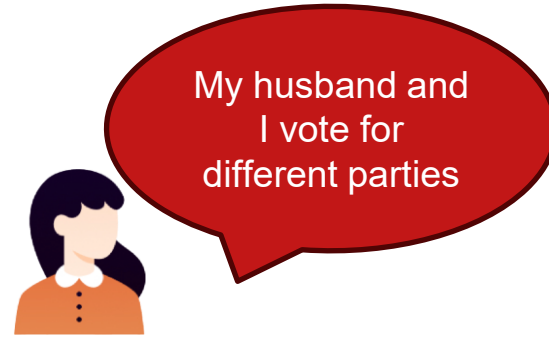
# ZSTS: Limitations

- **High computational cost.** Requires one LLM invocation per word in the document
- **Numeric substitutions break relationships.** Numbers are replaced with random digits of the same length, instead of BERT top-n predictions. Algebraic relationships are lost. For example: “*Born 1990, age 33 in 2023*” could be transformed into “*Born 4715, age 87 in 6204*”
- **Word level only.** The probability threshold flags individual unlikely words but cannot catch sensitive phrases made up of common words. For example: “*weekly chemotherapy*” both words are individually plausible in many contexts, but the combination is highly disclosive.



# Self-disclosure: Privacy leaks without PII

To express themselves, people often disclose sensitive information about themselves even while not revealing PII. How can we detect and soften such information?



No name, no address, but reveals **relationship status** and **gender** (implicitly) in a single sentence.

## Why this matters?

- Posts on social media are **public**, **permanent**, and **searchable**
- Scattered disclosures combine into one identity
- Real harm risk: profiling, doxxing, harassment





# Reducing privacy risks in online self-disclosure

Dou et al. [ACL'24] defined a taxonomy of 19 categories of self-disclosure split into demographic attributes and personal experiences

| Category                      | #Spans | Avg Len     | Example                                                                                     |
|-------------------------------|--------|-------------|---------------------------------------------------------------------------------------------|
| <b>Demographic Attributes</b> |        |             |                                                                                             |
| LOCATION                      | 525    | 5.70±3.85   | I live in the UK and a diagnosis is really expensive, even with health insurance            |
| AGE                           | 308    | 2.93±1.72   | I am a 23-year-old who is currently going through the last leg of undergraduate school      |
| RELATIONSHIP STATUS           | 287    | 6.72±5.97   | My partner has not helped at all, and I'm bed ridden now                                    |
| AGE/GENDER                    | 248    | 1.42±0.71   | For some context, I (20F), still live with my parents                                       |
| PET                           | 192    | 6.93±7.31   | Hi, I have two musk turtles and have never had any health problems before at all            |
| APPEARANCE                    | 173    | 6.96±6.25   | Same here. I am 6'2. No one can sit behind me.                                              |
| HUSBAND/BF                    | 148    | 6.89 ±7.24  | My husband and I vote for different parties                                                 |
| WIFE/GF                       | 144    | 5.24±4.42   | My gf and I applied, we're new but fairly active!                                           |
| GENDER                        | 110    | 3.28±3.10   | Am I insane? Eh. I'm just a girl who wants to look on the outside how I feel on the inside. |
| RACE/NATIONALITY              | 99     | 3.63±2.37   | As Italian I hope tonight you will won the world cup                                        |
| SEXUAL ORIENTATION            | 58     | 6.52±7.47   | I'm a straight man but I do wanna say this                                                  |
| NAME                          | 21     | 3.81±3.48   | Hello guys, my name is xxx and I love travelling                                            |
| CONTACT                       | 14     | 5.69±3.56   | xxx is my ig                                                                                |
| <b>Personal Experiences</b>   |        |             |                                                                                             |
| HEALTH                        | 783    | 10.36±9.78  | I am pretty sure I have autism, but I don't want to get an official diagnosis.              |
| FAMILY                        | 543    | 9.27±8.73   | My little brother (9M) is my pride and joy                                                  |
| OCCUPATION                    | 428    | 8.90±6.60   | I'm a motorcycle tourer (by profession), but when I'm off the saddle I'm mostly bored       |
| MENTAL HEALTH                 | 285    | 16.86±16.28 | I get asked this pretty regularly.. but I struggle with depression and ADHD                 |
| EDUCATION                     | 229    | 9.92±7.71   | Hi there, I got accepted to UCLA (IS), which I'm pumped about.                              |
| FINANCE                       | 153    | 12.00±9.19  | Yes. I was making \$68k a year and had around \$19k in debt                                 |

<https://aclanthology.org/2024.acl-long.741.pdf>



# Reducing privacy risks in online self-disclosure

- Based on the taxonomy, the authors create a human-annotated dataset with 2.4K Reddit posts covering 4.8K varied self disclosures.
- Every self disclosure *span* was labelled with context, not just the entity.

*“I live in the US”* ✓

*“US”* ✗

- NER already finds “US”, the leak is the framing that ties it to the user.



# Detecting self-disclosure spans

- RoBERTa and DeBERTaV3 were fine-tuned on the collected dataset to detect self-disclosures as a sequence tagging task. GPT-4 was prompted instead of finetuning.
- Performance for this task is measured using [partial span-F1](#): a predicted span is considered correct if the overlap with reference span is  $> 50\%$

| Class (#spans)        | RoBERTa      | DeBERTa      | GPT-4       |
|-----------------------|--------------|--------------|-------------|
| AGE (35)              | 72.46        | 70.77        | <b>80.0</b> |
| AGE&GENDER (17)       | <b>84.21</b> | 70.27        | 74.42       |
| RACE/NATIONALITY (8)  | 60.0         | <b>82.35</b> | 70.59       |
| GENDER (17)           | 61.11        | <b>72.73</b> | 57.14       |
| LOCATION (41)         | 71.26        | <b>73.33</b> | 54.35       |
| APPEARANCE (31)       | 64.41        | <b>67.74</b> | 42.55       |
| WIFE/GF (30)          | 66.67        | <b>75.86</b> | 64.52       |
| FINANCE (33)          | 68.66        | <b>71.43</b> | 54.55       |
| OCCUPATION (44)       | 64.44        | <b>65.22</b> | 52.75       |
| FAMILY (44)           | <b>58.70</b> | 49.02        | 58.25       |
| HEALTH (40)           | 56.84        | <b>58.82</b> | 38.02       |
| MENTAL HEALTH (46)    | <b>64.71</b> | 63.16        | 52.73       |
| HUSBAND/BF (14)       | <b>75.0</b>  | 70.59        | 68.97       |
| EDUCATION (21)        | 68.09        | <b>69.23</b> | 51.06       |
| PET (15)              | 46.15        | <b>55.17</b> | 48.28       |
| RELATION. STATUS (31) | 41.10        | <b>43.08</b> | 42.86       |
| SEXUAL ORIENT. (12)   | <b>76.19</b> | 58.33        | 69.57       |
| AVERAGE               | 64.71        | <b>65.71</b> | 57.68       |

Span detection is a boundary-sensitive tagging task, where fine-tuning smaller models still beats in-context prompting LLMs



# Abstracting self-disclosures

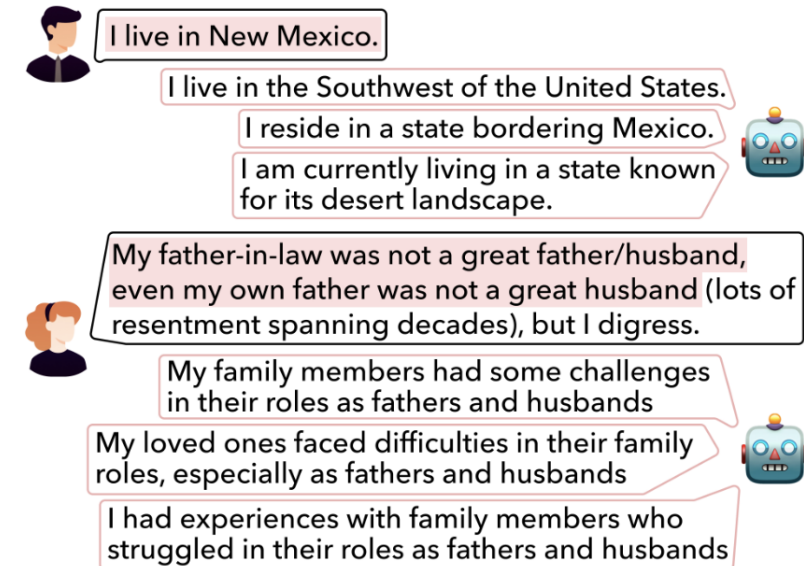
- After detecting self-disclosures, the goal is to re-write/abstract them to minimize privacy leakage.
- **Dataset:** GPT-3.5 with CoT prompting and few-shot demos are used on the earlier collected dataset to generate three diverse abstractions
- **Model:** Llama-2-7B fine-tuned with LoRA, producing three variants:
  1. **Sampling:** The model generates only one abstraction per input. Sampled thrice to get 3 abstractions.
  2. **End-to-end:** The model produces all three abstractions in a single pass
  3. **Iterative Generation:** The model produces each abstraction one at a time, each conditioned on the previous.
- Two different input formats are considered:

## SPECIAL TOKEN:

<SENTENCE>{s}<SPAN>{d}<ABSTRACTION>

## INSTRUCTION:

Your task is to abstract the given disclosure...  
Sentence:{s}\nDisclosure Span:{d}



# Abstracting self-disclosures: Results

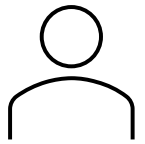
| Methods                     | Automatic Evaluation (Matching Metric) |              |              |              | Human Evaluation                                                                      |               |              |             |
|-----------------------------|----------------------------------------|--------------|--------------|--------------|---------------------------------------------------------------------------------------|---------------|--------------|-------------|
|                             | BLEU                                   | ROUGE-2      | ROUGE-L      | Diversity    | Rank Dist. (1→9)                                                                      | Rank $\nabla$ | Raw          | Z-Score     |
| <i>Sampling</i>             |                                        |              |              |              |                                                                                       |               |              |             |
| Special Token               | 17.40                                  | 18.81        | 38.42        | 2,576        |    | 5.37          | 79.22        | -0.10       |
| Instruction                 | 16.95                                  | 18.78        | 38.35        | 2,564        |    | 5.80          | 77.17        | -0.22       |
| Instruction (w/ thought)    | 17.06                                  | 18.47        | 38.43        | 2,882        |    | 5.32          | 78.13        | -0.13       |
| <i>End-to-end training</i>  |                                        |              |              |              |                                                                                       |               |              |             |
| Special Token               | 17.22                                  | 19.67        | 38.38        | 2,911        |    | 4.65          | 79.03        | -0.07       |
| Instruction                 | <u>17.99</u>                           | 19.60        | <b>39.57</b> | <u>2,992</u> |    | <b>3.77</b>   | <b>81.53</b> | <b>0.21</b> |
| Instruction (w/ thought)    | 16.53                                  | 19.24        | 38.81        | 2,801        |    | 4.48          | 79.76        | 0.04        |
| <i>Iterative generation</i> |                                        |              |              |              |                                                                                       |               |              |             |
| Special Token               | <b>18.13</b>                           | <u>19.74</u> | 38.71        | 2,913        |    | <u>4.12</u>   | 80.72        | 0.11        |
| Instruction                 | 17.80                                  | <b>19.81</b> | <u>39.56</u> | <b>3,067</b> |    | 5.10          | 78.46        | -0.03       |
| Instruction (w/ thought)    | 16.89                                  | 18.84        | 38.31        | 2,914        |  | 4.18          | <u>81.27</u> | <u>0.17</u> |

Rank Dist. shows how often each variant landed at each rank across all 9 (left = better) .  
 End-to-end + Instruction wins overall: best mean rank (3.77), raw (81.53), and z-score (+0.21)



# Inference: A new threat model

- An intelligent attacker can **infer** personal details from text that never appears to be a self-disclosure.



Kids today have Spotify, but they'll never know the stress of manually rewinding a cassette tape with a Nataraj pencil. Anyway, back to fighting with the HR portal to submit my investment proofs.

No date of birth. No age mentioned. No PII.  
But given the prompt, what can a LLM infer?

**Nationality:** Indian

**Age:** ~30-40 years old

**Occupation:** Salaried Corporate Employee

How the LLM connects the dots:

- **"Nataraj pencil" + "Cassette"**: Instantly identifies a distinctly Indian childhood lived during the 1990s.
- **"HR portal" + "investment proofs"**: Excludes business owners and freelancers; definitively tags the author as a salaried employee dealing with corporate tax compliance.
- **The Math**: A 90s kid working a corporate job today mathematically brackets the user in their thirties.

<https://arxiv.org/pdf/2310.07298>



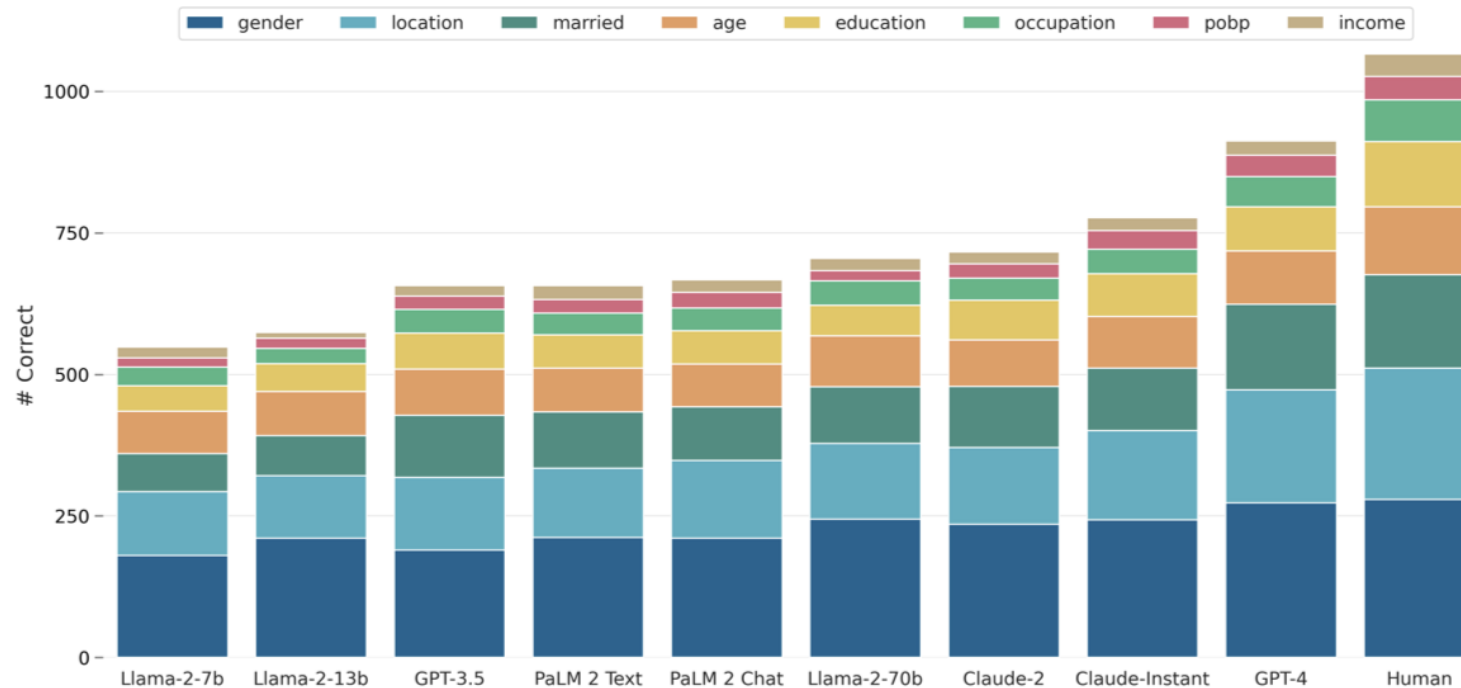
Trustworthy Large Language Models



Tanmoy Chakraborty

# LLMs as adversarial anonymizers

- LLM's strong reasoning capabilities make them powerful attribute inference attacker without any finetuning.
- [Staab et al. \[ICLR 2024\]](#) demonstrated that LLMs with just prompting can infer attributes such as age, occupation, income etc.



Attribute-inference accuracy on PersonalReddit:  
520 real Reddit profiles, human-labelled attributes

<https://arxiv.org/pdf/2402.13846>



# LLMs as adversarial anonymizers

The ability of LLMs to act as attribute inference attackers, can be used in an adversarial loop to find and remove the cues that leak them.

**Idea:** Use two frozen LLMs in a feedback loop [Staab et al., ICLR 2025]

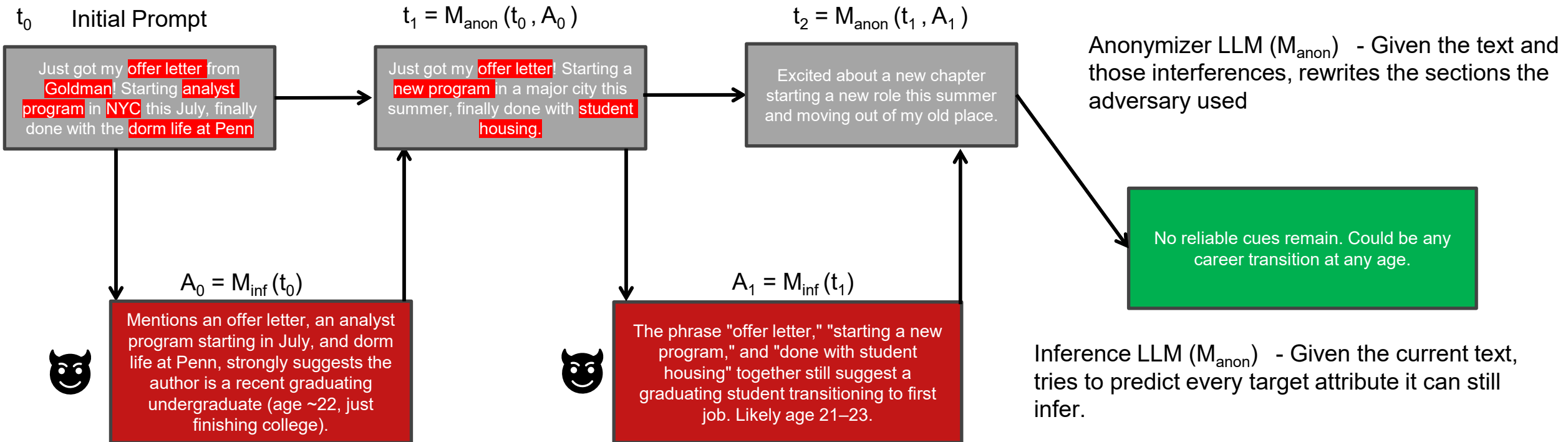
1. Inference model attempts to infer sensitive attribute and reports which cues gave it away
2. Anonymizer model uses the output of previous step as feedback and rewrites the text to remove those cues while preserving meaning
3. Repeat until the inference model is unable to guess any attributes or number of rounds are exhausted.

<https://arxiv.org/pdf/2402.13846>





# Adversarial anonymization loop



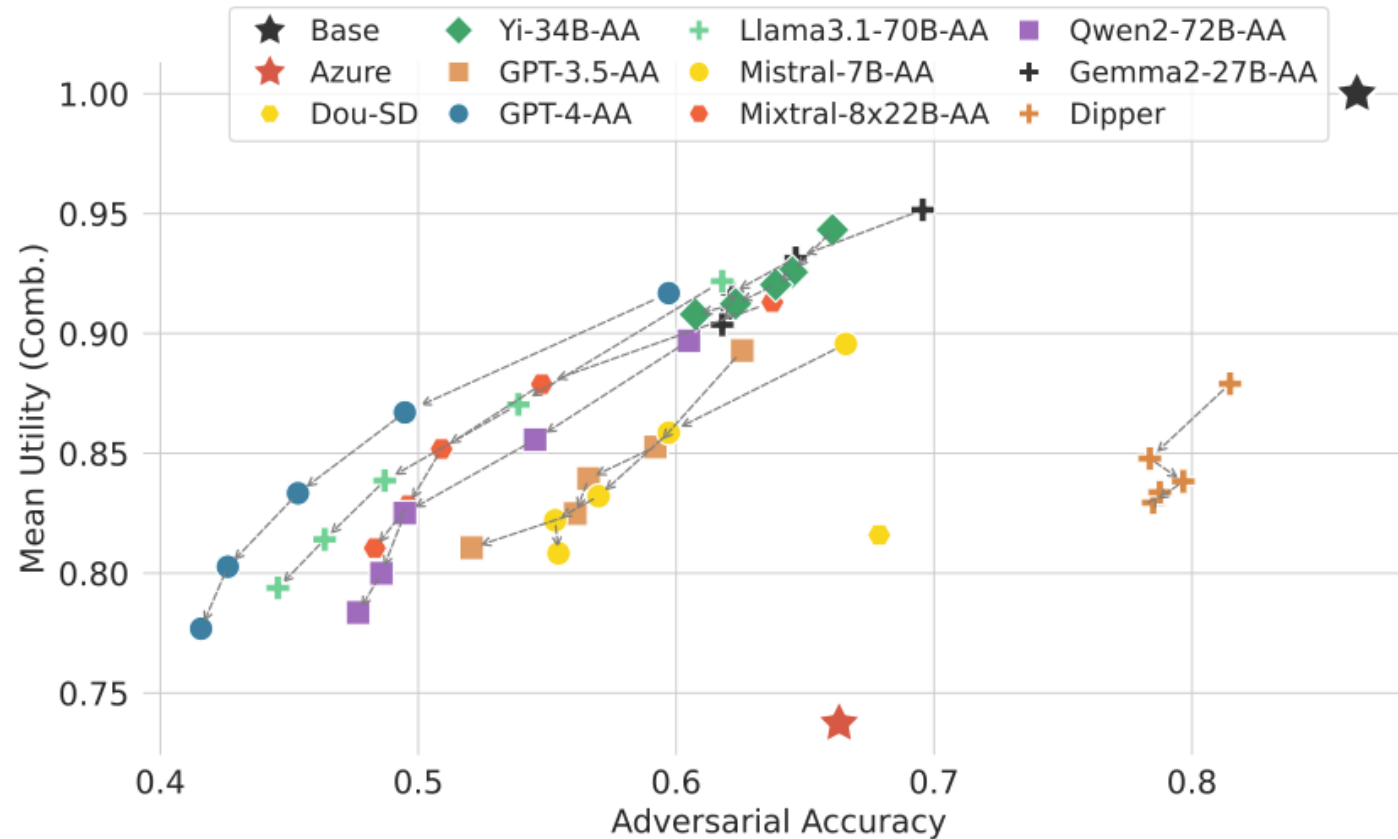
Each round, the adversary reveals what's still leaking. The anonymizer doesn't stop at named entities, it removes the structural cues too.



# Adversarial anonymization: Results

Utility is measured as the mean of:

1. GPT-4 utility judge score which considers readability + meaning
2. BLEU
3. ROUGE-1



Adversary LLM is able to predict the private attribute.



# Adversarial anonymization loop: Limitations

- Using two LLMs in the loop as inference attack and anonymizer removes the leaking cues → adversarial accuracy drops, privacy improves
- However, it also leads to a drop in utility, as utility score is measured but never optimized for.
- Utility isn't generic. It must be preserved for specific downstream tasks. Can we optimize anonymization for the utility that actually matters?



# Why task aware anonymization ?

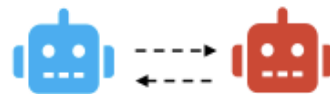
- Anonymization with adversarial feedback, checks only text quality.
- Utility for downstream tasks is never taken into consideration.
- Without downstream task specific utility, identity driven cuts destroy task specific details.

## Task: Occupation Classification



Original Document

Jacques "Toto" Brugnon (11 May 1895 – 20 March 1978) was a French tennis player, one of the famous "Four Musketeers" from France who dominated tennis in the late 1920s and early 1930s. He was born in Paris and died in Paris. He was primarily a doubles specialist who won 10 Grand Slam doubles titles in the French, American, Australian and British championships ...



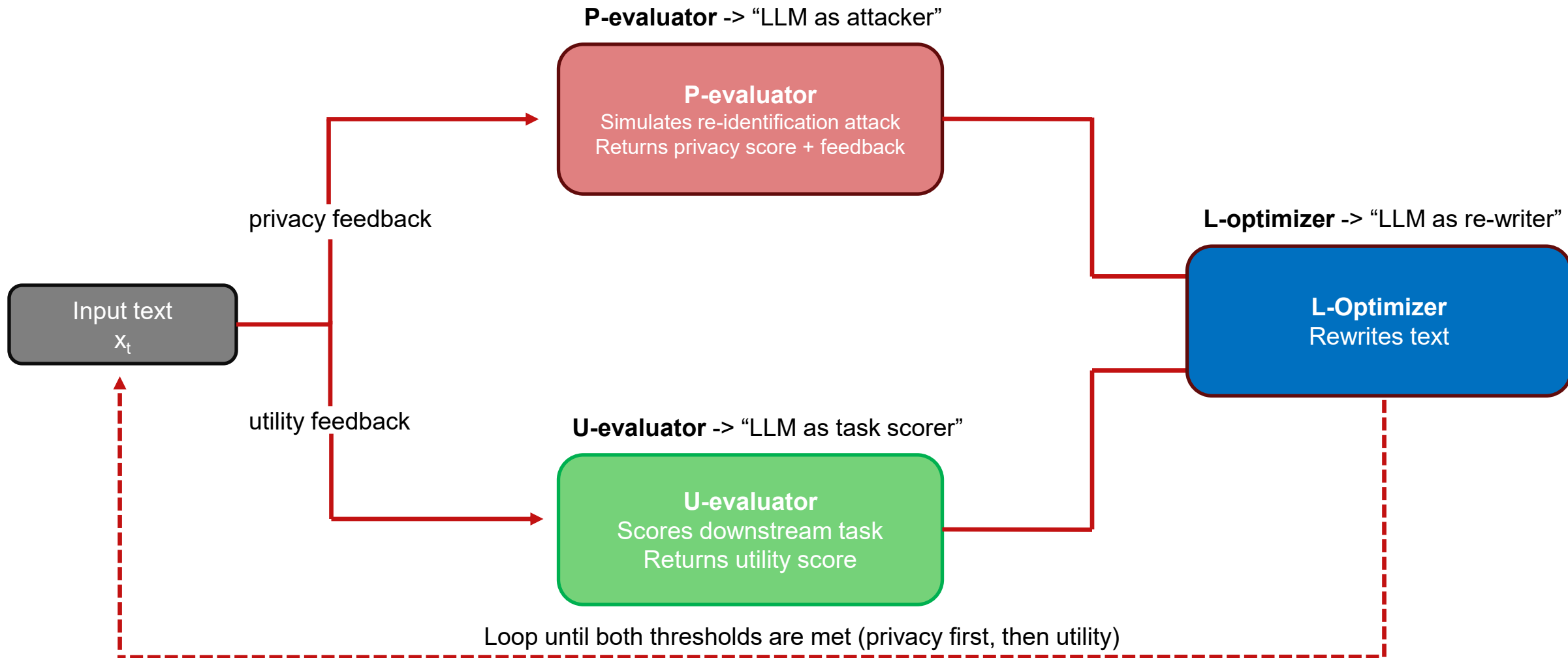
Adversarial Feedback

A person (born in a certain century – died in a certain century) was an athlete, one of a famous group from a certain region who dominated a sport in a certain era. He was born and died in his birth city. He was primarily a specialist who won many titles in various championships ...

<https://arxiv.org/pdf/2407.11770>



# RUPTA: Robust utility preserving text anonymization



# RUPTA: Lexicographic optimization

- RUPTA [Yang et al. ACL '25] frames the anonymization task as a lexicographic optimization problem
- Lexicographic optimization: A multi-objective optimization problem with multiple objectives, ranked by priority and solved in order
- Here, privacy is the primary goal, utility secondary. A rewrite is preferred if it has higher privacy, or equal privacy and higher utility
- While  $\text{privacy} < \text{max}$   $\rightarrow$  only anonymize
- Once  $\text{privacy} = \text{max}$   $\rightarrow$  raise utility without lowering privacy



# RUPTA: Scoring privacy and utility



Top-k identity guesses  $\rightarrow$  compare to ground truth identity  $y$  (*Jacques Burgnon*)

- $y$  in top-k  $\rightarrow p_t$  (privacy score) = it's rank
- $y$  not found  $\rightarrow p_t = K+1$  (maximum privacy)

$x_t$  + ground truth task label  $c$  (*tennis player*)  $\rightarrow$  score  $u_t$  (utility score)

- $u_t \rightarrow$  confidence that  $x_t$  still supports  $c$
- higher  $u_t \rightarrow$  more task signal preserved
- $K$  serves as the parameter that adjusts the sensitivity of the privacy evaluator. Higher  $K$  = stricter privacy
- U-evaluator can be instantiated as model used for downstream tasks, e.g. a sentiment classifier for sentiment analysis task



# RUPTA: Results

**Dataset:** DB-Bio, a collection of celebrity biographies from Wikipedia

|        | Method                            | Disclosure Risk |                           | Utility Preservation      |                           |                           |                           |                            |
|--------|-----------------------------------|-----------------|---------------------------|---------------------------|---------------------------|---------------------------|---------------------------|----------------------------|
|        |                                   | SR↓             | CS↓                       | Precision↑                | Recall↑                   | F1↑                       | Accuracy↑                 | Loss↓                      |
| DB-bio | Original                          | 100.00          | 98.45                     | 99.58                     | 99.68                     | 99.61                     | 99.58                     | 0.0422                     |
|        | Azure (Aahill, 2023)              | 78.24           | 80.87                     | 91.63                     | 95.04                     | 92.39                     | 92.47                     | 0.3202                     |
|        | DEID-GPT (Liu et al., 2023)       | 77.10           | 79.47                     | 90.82                     | 94.37                     | 92.56                     | 91.22                     | 0.3103                     |
|        | SD (Dou et al., 2023)             | 73.21           | 73.63                     | 92.27                     | 93.11                     | 92.69                     | 92.96                     | 0.2719                     |
|        | IncogniText (Frikha et al., 2024) | 58.06           | 56.28                     | 85.68                     | 89.03                     | 87.32                     | 88.28                     | 0.4842                     |
|        | AF (Staab et al., 2024b)          | <u>52.91</u>    | <b>50.84</b>              | 91.20                     | 94.26                     | 91.75                     | 92.02                     | 0.4048                     |
|        | RUPTA (Mixtral 8×22b)             | 67.78           | 67.15                     | <b>96.18</b>              | <b>97.13</b>              | <b>96.30</b>              | <b>96.23</b>              | <u>0.2167</u>              |
|        | RUPTA (Llama-3-70b)               | 64.02           | 63.23                     | 95.34                     | 96.23                     | 95.55                     | 95.82                     | 0.2224                     |
|        | RUPTA (GPT-3.5)                   | 68.51           | 69.16                     | 95.40                     | 96.02                     | 95.70                     | 95.49                     | 0.2188                     |
|        | RUPTA (GPT-4)                     | <b>52.67</b>    | <u>53.11</u> <sup>†</sup> | <u>95.58</u> <sup>†</sup> | <u>96.26</u> <sup>†</sup> | <u>95.91</u> <sup>†</sup> | <u>96.02</u> <sup>†</sup> | <b>0.1618</b> <sup>†</sup> |

SR => Success Rate in inferring identity from anonymized text

CS => Confidence Score when predicting ground truth personal information





# Summary: Text sanitization toolkit

| Method                     | Strength                           | Drawback                                       |
|----------------------------|------------------------------------|------------------------------------------------|
| NER / Rule-based           | Fast, well-understood              | Misses anything outside the entity list        |
| Zero-shot LLM substitution | Cross-domain, no-finetuning        | One LLM call per word; numeric relations break |
| Self-disclosure detection  | Catches leaks beyond PII           | Abstraction reduces utility; finite categories |
| Adversarial anonymization  | Robust against LLM attackers       | No utility check; cost scales with rounds      |
| RUPTA                      | Joint privacy-utility optimization | Three LLM calls per iteration; expensive       |

Each method targets a different threat, and pays a different cost



# Open challenges

- **No formal guarantee for free text:** no analog to  $k$  in  $k$ -anonymity exists for natural language.
- **Adversary capability scales with LLM size:** yesterday's anonymization is tomorrow's leak.
- **Latency and cost:** LLM-in-the-loop sanitization is expensive at scale.
- **Multilingual coverage:** most methods evaluated only in English; low resources languages remain underexplored.



Coming Next

# How LLMs leak

- Memorization and training-data extraction
- Membership inference attacks
- Prompt and in-context learning leakage
- Prompt injection



# How we defend

- Differential Privacy (DP)
- DP-based Stochastic Gradient Descent
- DP-based Text Sanitization

